

Paging

Some formulas to understand

Main memory related

Physical Address Space = Size of main memory

Size of main memory = Total number of frames x Page size

Frame size = Page size

If number of frames in main memory = 2^X , then number of bits in frame number = X bits

If Page size = 2^X Bytes, then number of bits in page offset = X bits

If size of main memory = 2^X Bytes, then number of bits in physical address = X bits

Process related

Virtual Address Space = Size of process

Number of pages the process is divided = Process size / Page size

If process size = 2^X bytes, then number of bits in virtual address space = X bits

Page table related

Size of page table = Number of entries in page table x Page table entry size

Number of entries in pages table = Number of pages the process is divided

Page table entry size = Number of bits in frame number + Number of bits used for optional fields if any

Others

In general, if the given address consists of 'n' bits, then using 'n' bits, 2^n locations are possible.

Then, size of memory = 2^n x Size of one location.

If the memory is byte-addressable, then size of one location = 1 byte.

Thus, size of memory = 2^n bytes.

If the memory is word-addressable where 1 word = m bytes, then size of one location = m bytes.

Thus, size of memory = 2^n x m bytes.

Sample Problems

1. Calculate the size of memory if its address consists of 22 bits and the memory is 2-byte addressable.

We have-

Number of locations possible with 22 bits = 2^{22} locations

It is given that the size of one location = 2 bytes

Thus, Size of memory = $2^{22} \times 2 \text{ bytes} = 2^{23} = 8 \text{ MB}$

2. Calculate the number of bits required in the address for memory having size of 16 GB. Assume the memory is 4-byte addressable.

Let 'n' numbers of bits are required. Then, Size of memory = $2^n \times 4 \text{ bytes}$.

Since, the given memory has size of 16 GB, so we have-

$$2^n \times 4 \text{ bytes} = 16 \text{ GB}$$

$$2^n \times 4 = 16 \text{ GB}$$

$$2^n \times 2^2 = 2^{34}$$

$$2^n = 2^{32}$$

$$n = 32 \text{ bits}$$

3. Consider a system with byte-addressable memory, 32 bit logical addresses, 4 kilobyte page size and page table entries of 4 bytes each. The size of the page table in the system in megabytes is _____.

Given-

Number of bits in logical address = 32 bits

Page size = 4KB

Page table entry size = 4 bytes

Number of bits in logical address = 32 bits

Thus, Process size = $2^{32} \text{ B} = 4 \text{ GB}$

Number of pages the process is divided = Process size / Page size = $4 \text{ GB} / 4 \text{ KB} = 2^{20}$ pages

Thus, number of entries in page table = 2^{20} entries

Page table size = Number of entries in page table x Page table entry size
 $= 2^{20} \times 4 \text{ bytes} = 4 \text{ MB}$

4. Consider a machine with 64 MB physical memory and a 32 bit virtual address space. If the page size is 4 KB, what is the approximate size of the page table?

Given

Size of main memory = 64 MB

Number of bits in virtual address space = 32 bits

Page size = 4 KB

We will consider that the memory is byte addressable.

Number of Bits in Physical Address

Size of main memory = $64 \text{ MB} = 2^{26} \text{ B}$

Thus, Number of bits in physical address = 26 bits

Number of frames in main memory = Size of main memory / Frame size
 $= 64 \text{ MB} / 4 \text{ KB} = 2^{26} \text{ B} / 2^{12} \text{ B} = 2^{14}$

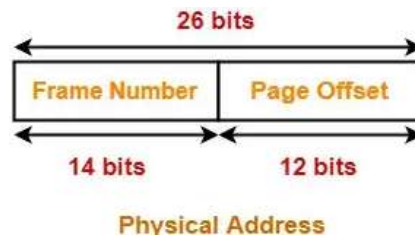
Thus, Number of bits in frame number = 14 bits

Number of Bits in Page Offset

We have, Page size = $4 \text{ KB} = 2^{12} \text{ B}$

Thus, Number of bits in page offset = 12 bits

So, Physical address is



Process Size

Number of bits in virtual address space = 32 bits

Thus, Process size = 2^{32} B = 4 GB

Number of Entries in Page Table

Number of pages the process is divided = Process size / Page size = 4 GB / 4 KB = 2^{20} pages

Thus, Number of entries in page table = 2^{20} entries

Page Table Size

Page table size = Number of entries in page table x Page table entry size

= Number of entries in page table x Number of bits in frame number

= $2^{20} \times 14$ bits = $2^{20} \times 16$ bits

= $2^{20} \times 2$ bytes = 2 MB

5. Consider a single level paging scheme. The virtual address space is 4 MB and page size is 4 KB. What is the maximum page table entry size possible such that the entire page table fits well in one page?

For page table, to fit well in one page, we must have-

Page table size $\leq$ Page size

Number of Pages of Process

Number of pages the process is divided = Process size / Page size = 4 MB / 4 KB = 2^{10} pages

Page Table Size

Let page table entry size = B bytes

Now,

Page table size = Number of entries in the page table x Page table entry size

= Number of pages the process is divided x Page table entry size = $2^{10} \times B$ bytes

Now,

According to the above condition, we must have $2^{10} \times B \leq 4 \text{ KB}$

$2^{10} \times B \leq 2^{12}$

$B \leq 4$

Thus, maximum page table entry size possible = **4 bytes**.

6. Consider a single level paging scheme. The virtual address space is 128 TB and page size is 32 MB. What is the maximum page table entry size possible such that the entire page table fits well in one page?

For page table, to fit well in one page, we must have Page table size $\leq$ Page size

Number of Pages of Process

Number of pages the process is divided = Process size / Page size = 128 TB / 32 MB
 $= 2^{47} \text{ B} / 2^{25} \text{ B} = 2^{22}$ pages

Page Table Size

Let page table entry size = B bytes

Now, Page table size = Number of entries in the page table x Page table entry size
 $= \text{Number of pages the process is divided} \times \text{Page table entry size} = 2^{22} \times \text{B bytes}$

Now, According to the above condition, we must have $2^{22} \times \text{B bytes} \leq 32 \text{ MB}$
 $2^{22} \times \text{B} \leq 2^{25}$

$\text{B} \leq 8$

Thus, maximum page table entry size possible = **8 bytes.**

7. Consider a system using paging scheme where

Logical Address Space = 4 GB

Physical Address Space = 16 TB

Page size = 4 KB

Find how many levels of page table will be required?

Number of Bits in Physical Address

Size of main memory = Physical Address Space = 16 TB = 2^{44} B

Thus, Number of bits in physical address = 44 bits

Number of Frames in Main Memory

Number of frames in main memory = Size of main memory / Frame size = 16 TB / 4 KB
 $= 2^{32}$ frames

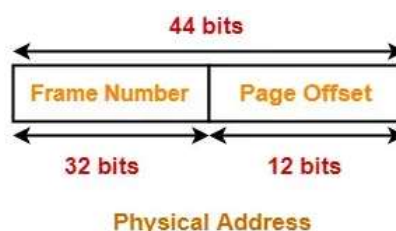
Thus, Number of bits in frame number = 32 bits

Number of Bits in Page Offset

We have, Page size = 4 KB = 2^{12} B

Thus, Number of bits in page offset = 12 bits

So, Physical address is



Number of Pages of Process

Number of pages the process is divided = Process size / Page size = 4 GB / 4 KB = 2^{20} pages

Inner Page Table Size

Inner page table keeps track of the frames storing the pages of process.

Inner Page table size = Number of entries in inner page table x Page table entry size

= Number of pages the process is divided x Number of bits in frame number

= $2^{20} \times 32$ bits

= $2^{20} \times 4$ bytes = 4 MB

Now, we can observe-

The size of inner page table is greater than the frame size (4 KB).

Thus, inner page table cannot be stored in a single frame.

So, inner page table has to be divided into pages.

Number of Pages of Inner Page Table

Number of pages the inner page table is divided = Inner page table size / Page size

= 4 MB / 4 KB = 2^{10} pages

Now, these 2^{10} pages of inner page table are stored in different frames of the main memory.

Number of Page Table Entries in One Page of Inner Page Table

Number of page table entries in one page of inner page table = Page size / Page table entry size

= Page size / Number of bits in frame number = 4 KB / 32 bits = 4 KB / 4 B = 2^{10}

Number of Bits Required to Search an Entry in One Page of Inner Page Table

One page of inner page table contains 210 entries.

Thus,

Number of bits required to search a particular entry in one page of inner page table = 10 bits

Outer Page Table Size

Outer page table is required to keep track of the frames storing the pages of inner page table.

Outer Page table size = Number of entries in outer page table x Page table entry size

= Number of pages the inner page table is divided x Number of bits in frame number

= $2^{10} \times 32$ bits = $2^{10} \times 4$ bytes = 4 KB

Now, we can observe-

The size of outer page table is same as frame size (4 KB).

Thus, outer page table can be stored in a single frame.

So, for given system, we will have two levels of page table.

Page Table Base Register (PTBR) will store the base address of the outer page table.

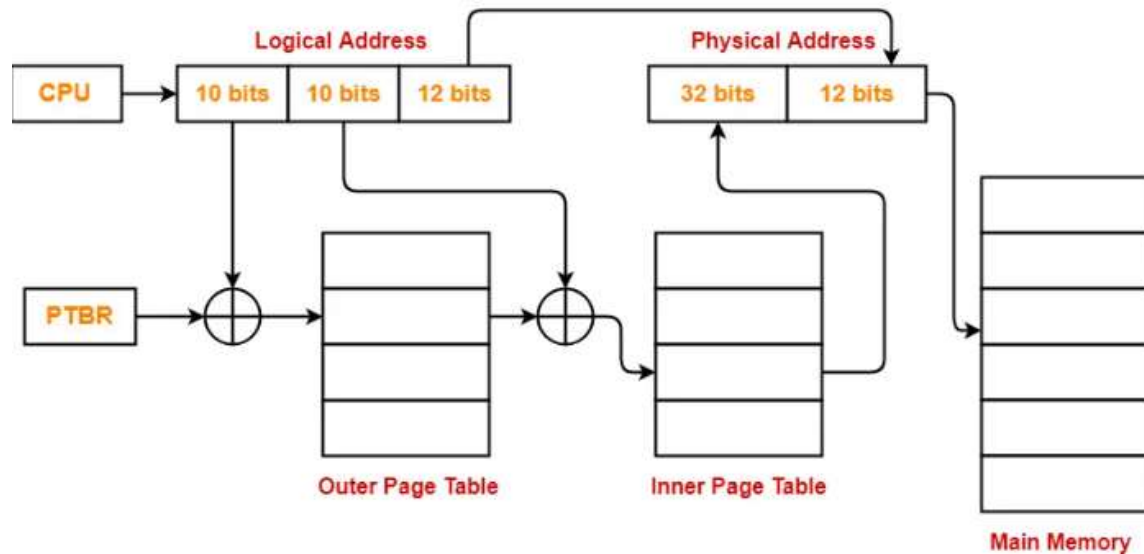
Number of Bits Required to Search an Entry in Outer Page Table

Outer page table contains 210 entries.

Thus,

Number of bits required to search a particular entry in outer page table = 10 bits

The paging system will look like as shown below:



8. Consider a system using multilevel paging scheme. The page size is 16 KB. The memory is byte addressable and virtual address is 48 bits long. The page table entry size is 4 bytes.

Find-

How many levels of page table will be required?

Give the divided physical address and virtual address.

Given-

Virtual Address = 48 bits

Page size = 16 KB

Page table entry size = 4 bytes

Number of Bits in Frame Number

We have, Page table entry size = 4 bytes = 32 bits

Thus, Number of bits in frame number = 32 bits

Number of Frames in Main Memory

We have, Number of bits in frame number = 32 bits

Thus, Number of frames in main memory = 2^{32} frames

Size of Main Memory

Size of main memory = Total number of frames x Frame size = $2^{32} \times 16 \text{ KB} = 2^{46} \text{ B} = 64 \text{ TB}$

Thus, Number of bits in physical address = 46 bits

Number of Bits in Page Offset

We have,

Page size = 16 KB = 2^{14} B

Thus, Number of bits in page offset = 14 bits

Process Size

Number of bits in virtual address = 48 bits

Thus, Process size = 2^{48} bytes = 256 TB

Number of Pages of Process

Number of pages the process is divided = Process size / Page size = 256 TB / 16 KB
= $2^{48} \text{ B} / 2^{14} \text{ B} = 2^{34}$ pages

Inner Page Table Size

Inner page table keeps track of the frames storing the pages of process.

Inner Page table size = Number of entries in inner page table x Page table entry size
= Number of pages the process is divided x Page table entry size
= $2^{34} \times 4 \text{ bytes} = 2^{36} \text{ bytes} = 64 \text{ GB}$

Now, we can observe-

The size of inner page table is greater than the frame size (4 KB).

Thus, inner page table cannot be stored in a single frame.

So, inner page table has to be divided into pages.

Number of Pages of Inner Page Table

Number of pages the inner page table is divided = Inner page table size / Page size
= 64 GB / 16 KB = $2^{36} \text{ B} / 2^{14} \text{ B} = 2^{22}$ pages

Now, these 2^{22} pages of inner page table are stored in different frames of the main memory.

Number of Page Table Entries in One Page of Inner Page Table

Number of page table entries in one page of inner page table = Page size / Page table entry size
= 16 KB / 4 B = 2^{12} entries

Number of Bits Required to Search an Entry in One Page of Inner Page Table

One page of inner page table contains 2^{12} entries.

Thus, Number of bits required to search a particular entry in one page of inner page table = 12 bits

Outer Page Table-1 Size

Outer page table-1 is required to keep track of the frames storing the pages of inner page table.

Outer Page table-1 size = Number of entries in outer page table-1 x Page table entry size
= Number of pages the inner page table is divided x Page table entry size
= $2^{22} \times 4 \text{ bytes} = 16 \text{ MB}$

Now, we can observe-

The size of outer page table-1 is greater than the frame size (4 KB).

Thus, outer page table-1 cannot be stored in a single frame.

So, outer page table-1 has to be divided into pages.

Number of Pages of Outer Page Table-1

Number of pages the outer page table-1 is divided = Outer page table-1 size / Page size
= 16 MB / 16 KB = 2^{10} pages

Now, these 2^{10} pages of outer page table-1 are stored in different frames of the main memory.

Number of Page Table Entries in One Page of Outer Page Table-1

Number of page table entries in one page of outer page table-1 = Page size / Page table entry size

$$= 16 \text{ KB} / 4 \text{ B} = 2^{12} \text{ entries}$$

Number of Bits Required to Search an Entry in One Page of Outer Page Table-1

One page of outer page table-1 contains 212 entries.

Thus, Number of bits required to search a particular entry in one page of outer page table-1 = 12 bits

Outer Page Table-2 Size-

Outer page table-2 is required to keep track of the frames storing the pages of outer page table-1.

Outer Page table-2 size = Number of entries in outer page table-2 x Page table entry size

= Number of pages the outer page table-1 is divided x Page table entry size

$$= 2^{10} \times 4 \text{ bytes} = 4 \text{ KB}$$

Now, we can observe-

The size of outer page table-2 is less than the frame size (16 KB).

Thus, outer page table-2 can be stored in a single frame.

In fact, outer page table-2 will not completely occupy one frame and some space will remain vacant.

So, for given system, we will have three levels of page table.

Page Table Base Register (PTBR) will store the base address of the outer page table-2.

Outer page table-2 contains 2^{10} entries.

Thus, Number of bits required to search a particular entry in outer page table-2 = 10 bits

The paging system will look like as shown below-

